# Data Communication & Computer Networks (CN)

## Topic 7 of 7 : Network Security

**0 MCQs** — Repo Exam Questions + Forouzan — bilingual (English + Hindi), exam-ready

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • RSMSSB IA • RSMSSB SA • Data Entry/DD • BCI • SCI • REET • RAS & other Rajasthan Govt. Exams**

Total Questions: **0** MCQs

Question Type: MCQ — 4 options (A, B, C, D) per question, in English + Hindi

Layout: Left = **question** | Right = **explanation** (green): ✔ **why the correct answer is right** + ★ **Nukta** (key exam point)

Language rule: Hindi is the second language; **technical terms stay in English** (topology, protocol, jitter…)

Verification: Every answer cross-verified **4×** (Forouzan + official answer keys of Subject Test / Unit Test Solutions). Wrong options verified internally.

## Syllabus Coverage — sub-topic wise order

**How to read:** Left side: question + 4 options (try it yourself first). Right side (green): ✔ correct answer with why it is right, and ★ **Nukta** — the one-line exam tip to remember. Practice in 2 passes: first read question + options only, then check the green column.

**System:** generated by **tools/cn_build_topic.py** from **tools/data/topic7_data.py**. To add more questions later: drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions auto-classify topic-wise. See **Computer Network/README.md**.

## Pending Review — auto-ingested, needs verification

Auto-extracted & classified; answers to be verified before counting.

**P1.** Network Attacks & Description 9 List–I (Attack
**(A)** Phishing ( )
**(B)** DoS Attack
**(C)** Man-in-the-Middle
**(D)** Spoofing (R ) List–II (Description / ) (i) Flooding network traffic / 9 (ii) Stealing user credentials (iii) Intercepting communication B W+ (iv) Impersonation attack Codes / : (A) a–ii, b–i, c–iii, d–iv (B) a–i, b–ii, c–iii, d–iv (C) a–iii, b–i, c–ii, d–iv (D) a–ii, b–iii, c–i, d–iv

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

**P2.** In the digital signature technique when the whole message is signed using an asymmetric key, the receiver of the message uses _____ to verify the signature. v B Asymmetric Key W + , - y W + z _____

**(A)** Her or his own public key
**(B)** Her or his own private key
**(C)** The sender’s public key
**(D)** None of these

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P3.** The text that gets transformed using an algorithm cipher is called: " $ 2 $ : ;

**(A)** Complex Text
**(B)** Transformed Text
**(C)** Plain Text
**(D)** Scalar Text

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P4.** In cryptography, which statement about Plaintext, Ciphertext and Key is MOST accurate? ‰Š ‹ B \ Œ, Œ

**(A)** Plaintext is generated after decryption using a public key
**(B)** Ciphertext is the original readable message
**(C)** Plaintext is transformed into Ciphertext using an algorithm and a key
**(D)** Ciphertext and plaintext are always of the same length

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P5.** Which attack becomes possible if the same key is reused multiple times in symmetric encryption

**(A)** Replay Attack
**(B)** Known Plaintext Attack
**(C)** Man-in-the-Middle Attack
**(D)** Phishing Attack

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

**P6.** Data Encryption Standard (DES) ; ?Data Encryption Standard (DES) is a:

**(A)** Block Cipher
**(B)** Stream Cipher
**(C)** Bit Cipher
**(D)** Byte Cipher

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P7.** In the virus life cycle, which phase actually performs the malicious activity? H B - h

**(A)** Infection Phase / Z
**(B)** Triggering Phase
**(C)** Dormant Phase / C
**(D)** Execution (Payload) Phase / F

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

**P8.** Which of the following backup media provides Write Once Read Many (WORM) functionality? k l B Write Once Read Many (WORM) @

**(A)** Hard Disk / h 1
**(B)** Magnetic Tape / j
**(C)** Optical Disk (CD-R / DVD-R)
**(D)** Flash Drive / k 9

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

**P9.** IPSec operates in which mode(s)? IPSec B P

**(A)** Transport mode only / 9 h
**(B)** Tunnel mode only
**(C)** Transport mode and Tunnel mode / 9 h
**(D)** Application mode / n$

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IPSec & Security Protocols]

⚠ auto-ingested — answer needs verification

**P10.** SSL/TLS handshake protocol is mainly used for:SSL/TLS u @ v w x ;

**(A)** Data encryption
**(B)** Data compression
**(C)** Key exchange and authentication
**(D)** Packet routing

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IPSec & Security Protocols]

⚠ auto-ingested — answer needs verification

**P11.** IPSec is MOST commonly used to implement: IPSec

**(A)** Secure Email
**(B)** Secure Web Browsing
**(C)** Virtual Private Network (VPN)
**(D)** Wireless Security

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

**P12.** In IPSec Tunnel mode:/ IPSec B ;

**(A)** Only payload is encrypted
**(B)** Only header is encrypted
**(C)** Entire IP packet is encrypted
**(D)** No encryption is used

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IPSec & Security Protocols]

⚠ auto-ingested — answer needs verification

**P13.** Which of the following uses the same key for encryption and decryption? H I + - I}y~ y~

**(A)** Symmetric Key Cipher
**(B)** Asymmetric Key Cipher
**(C)** Hash Function
**(D)** Digital Signature

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P14.** Which of the following uses a pair of keys (public and private)? H I + ( 9 )

**(A)** Symmetric Cipher
**(B)** Asymmetric Cipher
**(C)** DES
**(D)** AES

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P15.** Which of the following is faster in encryption and decryption? I}y~ y~ + -

**(A)** Symmetric Key Cipher
**(B)** Asymmetric Key Cipher
**(C)** Can not decide
**(D)** None of the these

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P16.** Public key is used to _____. I• _____

**(A)** Encrypt the data
**(B)** Decrypt the data
**(C)** Both encrypt and decrypt
**(D)** Generate session keys

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P17.** Which of the following is NOT a symmetric encryption algorithm? H I I}y~ € d •

**(A)** AES
**(B)** DES
**(C)** Blowfish
**(D)** RSA

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P18.** What is the main disadvantage of symmetric key encryption? h I}y~ T

**(A)** It is very slow
**(B)** Key distribution is difficult
**(C)** It uses two keys
**(D)** It cannot encrypt text

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P19.** In public key cryptography, the message encrypted with the public key can only be decrypted by _____. I• y, m +, I• I}y, _____ ƒ

**(A)** Another public key
**(B)** Same public key
**(C)** Private key corresponding to that public key
**(D)** Session key

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P20.** Which one of the following statements is true about symmetric and asymmetric cryptography? h h y, m + H + -

**(A)** Symmetric cryptography provides authentication.
**(B)** Asymmetric cryptography is faster than symmetri
**(C)** C. Symmetric cryptography requires secure key exchange.
**(D)** Both use two different keys.

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P21.** What is the key length of AES-256 encryption? AES-256 I}y~

**(A)** 128 bits
**(B)** 192 bits
**(C)** 256 bits
**(D)** 512 bits

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P22.** Which cipher method encrypts one bit or byte at a time? I}y,

**(A)** Block Cipher
**(B)** Stream Cipher
**(C)** Substitution Cipher
**(D)** Transposition Cipher

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P23.** Which of the following statements is FALSE? H + -

**(A)** AES is a block cipher.
**(B)** RSA is a symmetric cipher.
**(C)** Diffie-Hellman is used for key exchange.
**(D)** DES is symmetric. Which of the following is NOT a goal of network security? H I + - 9 t „ ? A. Confidentiality ( ) B. Integrity ( ) C. Availability ( Y ) D. Redundancy ( )

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P24.** The term "CIA triad" in information security refers to _____. t + "CIA ] " _____
**(A)** Cryptography, Integrity, Authentication
**(B)** Confidentiality, Integrity, Availability
**(C)** Confidentiality, Information, Access
**(D)** Control, Information, Access
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P25.** The goal of “Integrity” ensures that
**(A)** Data can be accessed only by authorized users
**(B)** Data is accurate and has not been modified
**(C)** Data is always available when needed
**(D)** Unauthorized users cannot enter the system 40 Which of the following attack violates confidentiality? H I + j ? A. Interception B. Modification C. Fabrication D. Interruption
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P26.** Modification attack violates which security goal? t „ j
**(A)** Confidentiality
**(B)** Integrity
**(C)** Availability
**(D)** Authentication
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P27.** Denial of Service (DoS) attack violates which goal? Denial of Service (DoS) „ j
**(A)** Confidentiality
**(B)** Integrity
**(C)** Availability
**(D)** Authentication
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

**P28.** Spoofing attack targets which type of security aspect? ˆ t t
**(A)** Authentication
**(B)** Integrity
**(C)** Availability
**(D)** Confidentiality
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P29.** Which of the following is an active attack? H I + - y (Active)
**(A)** Interception
**(B)** Modification
**(C)** Traffic Analysis
**(D)** Eavesdropping
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

**P30.** The Caesar Cipher is an example of which type of cryptographic technique? Caesar Cipher y, m

**(A)** Transposition Cipher
**(B)** Substitution Cipher
**(C)** Stream Cipher
**(D)** Block Cipher In Steganography, the file used to hide secret data is called _____. Steganography + + p A. Secret File B. Cover File C. Cipher File D. Key File

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P31.** The process of detecting hidden data inside a file is known as _____. ? y

**(A)** Cryptanalysis
**(B)** Steganalysis
**(C)** Data Mining
**(D)** Stegnography

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P32.** What is the output of a Steganography process called? Steganography ? y

**(A)** Ciphertext
**(B)** Stego-object
**(C)** Encoded file
**(D)** Key file

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

**P33.** Which of the following is used in HTTPS for encryption?HTTPS

**(A)** SSL/TLS
**(B)** FTP
**(C)** ARP
**(D)** NAT

[Source: CN Topic Test 5.pdf | auto-classified: Topic 7 / IPSec & Security Protocols]

⚠ auto-ingested — answer needs verification

**P34.** What type of NAT maps multiple private IPs to one public IP? NAT IP IP

**(A)** Static NAT
**(B)** Dynamic NAT
**(C)** PAT (Port Address Translation)
**(D)** Double NAT

[Source: CN Topic Test 5.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

**P35.** Which mode of IPsec provides end-to- end security?IPsec

**(A)** Transport mode
**(B)** Tunnel mode
**(C)** Static mode
**(D)** Gateway mode What type of attack is a Smurf attack?Smurf (a) DoS (b) Spoofing (c) Phishing (d) Brute force

[Source: CN Topic Test 5.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

| | |
|---|---|
| **P36.** What is the function of a proxy server<br>**(A)** Asssign IP address<br>**(B)** Direct connect to the internet<br>**(C)** Act as an intermediary between client and server<br>**(D)** Block all network traffic<br>[Source: CN Topic Test 5.pdf \| auto-classified: Topic 1 / Types of Networks] | ⚠ auto-ingested — answer needs verification |
| **P37.** What is the advantage of setting up a DMZ (Demilitarized Zone) using firewall?<br>**(A)** To create multiple zones of trust<br>**(B)** Load balancing on the server<br>**(C)** Improve network performance<br>**(D)** Brings internet service into home from internet service provider<br>[Source: RPSC Programmer 2024 Paper-1 (with official key) \| auto-classified: Topic 1 / Types of Networks] | ⚠ auto-ingested — answer needs verification |